1. MC.02.01.ALGO (Algorithmic)

Accumulating costs means that

a. costs must be expensed in the period incurred.

b. costs must be measured and tracked.

c. costs must be compared to budgeted costs.

d. costs must be summed and entered on the income statement.

e. costs must be compared to previous periods to identify trends.

Feedback

Check My Work

Review meaning of accumulating costs.

Solution

b

2. MC.02.02.ALGO (Algorithmic)

Product (or manufacturing) costs consist of

a. direct materials, direct labor, manufacturing overhead, and operating expense.

b. manufacturing overhead, indirect labor and indirect materials.

c. direct materials, direct labor, and selling costs.

d. manufacturing overhead, direct materials and supplies expense.

e. prime costs and manufacturing overhead.

Feedback

Check My Work

Prime costs consist of direct materials and direct labor. Product costs consist of all costs used for the production of the product.

Solution

e

3. MC.02.06

Wachman Company produces a product with the following per-unit costs:

Direct materials	$15
Direct labor	6
Manufacturing overhead	19

Last year, Wachman produced and sold 2,000 units at a price of $75 each. Total selling and administrative expense was $30,000.

Conversion cost per unit was

a. $21.

b. $25.

c. $34.

d. $40.

e. None of these choices are correct.

Feedback

Check My Work

Conversion cost is the sum of direct labor and manufacturing overhead.

Post-Submission

Conversion Cost per Unit = $6 + $19 = $25

Solution
b

4. MC.02.07

Wachman Company produces a product with the following per-unit costs:

Direct materials	$15
Direct labor	6
Manufacturing overhead	19

Last year, Wachman produced and sold 2,000 units at a price of $75 each. Total selling and administrative expense was $30,000.

Total gross margin for last year was

a. $40,000.

✓ b. $70,000.

c. $80,000.

d. $88,000.

e. $100,000.

Feedback

Check My Work

Sales minus cost of goods sold = gross margin.
Multiply product cost by the number of units produced.

Post-Submission

Sales = $75 × 2,000 units = $150,000

Production Cost per Unit = $15 + $6 + $19 = $40

Cost of Goods Sold = $40 × 2,000 = $80,000

Gross Margin = $150,000 − $80,000 = $70,000

Solution
b

5. MC.02.08.ALGO (Algorithmic)

The accountant in a factory that produces biscuits for fast-food restaurants wants to assign costs to boxes of biscuits. Which of the following costs can be traced directly to boxes of biscuits?

a. The wages of the mixing labor

b. The cost of the boxes

c. The cost of milk and eggs used in biscuit mix

d. The cost of flour and baking soda

✓ e. All of these

Feedback

Check My Work

Direct costs can be traced directly to the cost object and are costs necessary for the completion of the activity.

Solution
e

6. MC.02.09.ALGO (Algorithmic)

Which of the following is an indirect cost?

a. The cost of mixing labor in a factory that makes over-the-counter pain relievers

✓ b. The cost of restriping the parking lot at a perfume factory

c. The cost of bottles in a shampoo factory

d. The cost of denim in a jeans factory

e. None of these choices are correct

Feedback

Check My Work

Indirect costs are not directly traced to a cost object.

Solution
b

7. MC.02.10.ALGO (Algorithmic)

Bobby Dee's is an owner-operated company that details (thoroughly cleans—inside and out) automobiles. Bobby Dee's is which of the following?

a. Manufacturing firm

b. Management firm

c. Wholesaler

✓ d. Service firm

e. All of these choices are correct

Feedback

Check My Work

Manufacturing companies produce products and have inventory accounts. Retailers and wholesalers buy products as inventory and then sell products for profit. Service companies provide intangible products/services. Services do not use an inventory account.

Solution
d

8. MC.02.11

Kellogg's makes a variety of breakfast cereals. Kellogg's is which of the following?

a. Wholesaler

b. Retailer

c. Service firm

✓ d. Manufacturing firm

e. None of these

Feedback

Check My Work

Manufacturing companies produce products and have inventory accounts. Retailers and wholesalers buy products as inventory and then sell products for profit. Service companies provide intangible products/services. Services do not use an inventory account.

Solution
d

9. MC.02.12

Target is which of the following?

- a. Wholesaler
- ✓ b. Retailer
- c. Service firm
- d. Manufacturing firm
- e. None of these

Feedback

Check My Work

Manufacturing companies produce products and have inventory accounts. Retailers and wholesalers buy products as inventory and then sell products for profit. Service companies provide intangible products/services. Services do not use an inventory account.

Solution
b

10. MC.02.14

JackMan Company produces die-cast metal bulldozers for toy shops. JackMan estimated the following average costs per bulldozer:

Direct materials	$8.65
Direct labor	1.10
Manufacturing overhead	0.95

Prime cost per unit is

- a. $8.65.
- b. $1.10.
- c. $0.95.
- d. $2.05.
- ✓ e. $9.75.

Feedback

Check My Work

Prime costs consist of direct materials and direct labor.

Post-Submission

Prime Cost per Unit = $8.65 + $1.10 = $9.75

Solution
e

11. MC.02.15

Which of the following is a period expense?

a. Factory insurance

b. CEO salary

c. Direct labor

d. Factory maintenance

e. All of these

Feedback

Check My Work

Period costs are other costs necessary in operations of the company, but not directly related to production, thus not carried as an inventory cost.

Solution
b

12. MC.02.17

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000
Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

Prime cost per unit is

a. $7.00.

b. $20.00.

c. $15.00.

d. $5.00.

e. $27.60.

Feedback

Check My Work

Prime costs consist of direct materials and direct labor. Compute prime cost. Then divide by number of units.

Post-Submission

Total Prime Cost = $50,000 + $20,000 = $70,000

Prime Cost per Unit = $70,000/10,000 units = $7.00

Solution
a

13. MC.02.18

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000

Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

Conversion cost per unit is

a. $7.00.

b. $20.00.

✓ c. $15.00.

d. $5.00.

e. $27.60.

Feedback

Check My Work

Conversion cost is the sum of direct labor and manufacturing overhead. Compute conversion cost. Then divide by number of units.

Post-Submission

Total Conversion Cost = $20,000 + $130,000 = $150,000

Conversion Cost per Unit = $150,000/10,000 units = $15.00

Solution

c

14. MC.02.19

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000
Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

Refer to the information for Barnard Company above. The cost of goods sold per unit is

a. $7.00.

✓ b. $20.00.

c. $15.00.

d. $5.00.

e. $27.60.

Feedback

Check My Work

Add the components of cost of goods sold and then divide by the number of units.

Post-Submission

Cost of Goods Sold = $50,000 + $20,000 + $130,000 = $200,000

Cost of Goods Sold per Unit = $200,000/10,000 units = $20.00

Solution
b

15. MC.02.20

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000
Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

The gross margin per unit is

a. $24.00.

✓ b. $11.00.

c. $16.00.

d. $26.00.

e. $3.40.

Feedback

Check My Work

Sales minus cost of goods sold = gross margin. Divide gross margin by the number of units sold.

Post-Submission

Sales = $31 × 10,000 = $310,000

Gross Margin = $310,000 – $200,000 = $110,000

Gross Margin per Unit = $110,000/10,000 units = $11.00

Solution
b

16. MC.02.21

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000
Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

The total period expense is

a. $276,000.

b. $200,000.

✓ c. $76,000.

d. $40,000.

e. $36,000.

Feedback

Check My Work

Period costs are other costs necessary in operations of the company, but not directly related to production, thus not carried as an inventory cost.

Post-Submission

Period Expense = $40,000 + $36,000 = $76,000

Solution

c

17. MC.02.22

Last year, Barnard Company incurred the following costs:

Direct materials	$ 50,000
Direct labor	20,000
Manufacturing overhead	130,000
Selling expense	40,000
Administrative expense	36,000

Barnard produced and sold 10,000 units at a price of $31 each.

Operating income is

✓ a. $34,000.

b. $110,000.

c. $234,000.

d. $270,000.

e. $74,000.

Feedback

Check My Work

Sales minus cost of goods sold = gross margin, and then subtract the period costs for operating income.

Post-Submission

Operating Income = $310,000 − $200,000 − $76,000 = $34,000

Solution

a

18. EX.02.48.ALGO (Algorithmic)

Total and Unit Product Cost

Martinez Manufacturing Inc. showed the following costs for last month:

Direct materials	$8,000
Direct labor	2,200
Manufacturing overhead	2,100
Selling expense	8,000

Last month, 3,000 units were produced and sold.

Required:

1. Classify each of the costs as product cost or period cost.

Direct materials	Product cost ✓
Direct labor	Product cost ✓
Manufacturing overhead	Product cost ✓
Selling expense	Period cost ✓

2. What is the total product cost for last month?

$ ✓

12300

3. What is the unit product cost for last month? **If required, round your answer to the nearest cent.**

$ ✓

4.1

per unit

Feedback

Check My Work

1. Period costs are other costs necessary in operations of the company, but not directly related to production, thus not carried as an inventory cost. Product costs consist of all costs used for the production of the product.

2. Add product costs for the total product cost.

3. Divide total product costs by the number of units produced.

Post-Submission

2.

Direct materials	$ 8,000
Direct labor	2,200
Manufacturing overhead	2,100
Total product cost	$12,300

3. Unit Product Cost = $12,300/3,000 units = $4.10

Solution

Total and Unit Product Cost

Martinez Manufacturing Inc. showed the following costs for last month:

Direct materials	$8,000

Direct labor	2,200
Manufacturing overhead	2,100
Selling expense	8,000

Last month, 3,000 units were produced and sold.

Required:

1. Classify each of the costs as product cost or period cost.

Direct materials	Product cost
Direct labor	Product cost
Manufacturing overhead	Product cost
Selling expense	Period cost

2. What is the total product cost for last month?

$ 12300

3. What is the unit product cost for last month? **If required, round your answer to the nearest cent.**

$ 4.1 per unit

19. EX.02.50

Classifying Cost of Production

A factory manufactures jelly. The jars of jelly are packed six to a box, and the boxes are sold to grocery stores. The following types of cost were incurred:

Jars
Sugar
Fruit
Pectin (thickener used in jams and jellies)
Boxes
Depreciation on the factory building
Cooking equipment operators' wages
Filling equipment operators' wages
Packers' wages
Janitors' wages
Receptionist's wages
Telephone
Utilities
Rental of Santa Claus suit (for Christmas party for children of factory workers)
Supervisory labor salaries
Insurance on factory building
Depreciation on factory equipment
Oil to lubricate filling equipment

Required:

Classify each of the costs as direct materials, direct labor, or overhead. The row for "Jars" is filled in as an example.

Costs	**Classification**
Jars	Direct Materials
Sugar	**Direct Materials** ✓
Fruit	**Direct Materials** ✓

Pectin	Direct Materials
Boxes	Direct Materials
Depreciation on the factory building	Overhead
Cooking equipment operators' wages	Direct Labor
Filling equipment operators' wages	Direct Labor
Packers' wages	Direct Labor
Janitors' wages	Overhead
Receptionist's wages	Overhead
Telephone	Overhead
Utilities	Overhead
Rental of Santa Claus suit	Overhead
Supervisory labor salaries	Overhead
Insurance on factory building	Overhead
Depreciation on factory equipment	Overhead
Oil to lubricate filling equipment	Overhead

Feedback

Check My Work

Direct materials are those materials that are a part of the final product and can be directly traced to the goods being produced.

Direct labor is the labor that can be directly traced to the goods being produced.

All product costs other than direct materials and direct labor are put into a category called manufacturing overhead.

Solution

Classifying Cost of Production

A factory manufactures jelly. The jars of jelly are packed six to a box, and the boxes are sold to grocery stores. The following types of cost were incurred:

Jars	Receptionist's wages
Sugar	Telephone
Fruit	Utilities
Pectin (thickener used in jams and jellies)	Rental of Santa Claus suit (for Christmas party for children of factory workers)
Boxes	Supervisory labor salaries
Depreciation on the factory building	Insurance on factory building
Cooking equipment operators' wages	Depreciation on factory equipment
Filling equipment operators' wages	Oil to lubricate filling equipment
Packers' wages	
Janitors' wages	

Janitors' wages

Required:

Classify each of the costs as direct materials, direct labor, or overhead. The row for "Jars" is filled in as an example.

Costs	Classification
Jars	Direct Materials
Sugar	Direct Materials
Fruit	Direct Materials
Pectin	Direct Materials
Boxes	Direct Materials
Depreciation on the factory building	Overhead
Cooking equipment operators' wages	Direct Labor
Filling equipment operators' wages	Direct Labor
Packers' wages	Direct Labor
Janitors' wages	Overhead
Receptionist's wages	Overhead
Telephone	Overhead
Utilities	Overhead
Rental of Santa Claus suit	Overhead
Supervisory labor salaries	Overhead
Insurance on factory building	Overhead
Depreciation on factory equipment	Overhead
Oil to lubricate filling equipment	Overhead

20. EX.02.51.ALGO (Algorithmic)

Product Cost in Total and Per Unit

Grin Company manufactures digital cameras. In January, Grin produced 2,750 cameras with the following costs:

Direct materials	$840,000
Direct labor	168,000
Manufacturing overhead	672,000

There were no beginning or ending inventories of WIP.

Required:

1. What was the total product cost in January?

$ ✓

1680000

2. What was the product cost per unit in January? **Round your answer to the nearest cent.**

$ ✓

610.91

per unit

Feedback

Check My Work

1. Total product cost is the total of direct and indirect costs associated with producing or acquiring a product and preparing it for sale.

2. Divide total product costs by the number of units produced.

Post-Submission

1.

Direct materials	$ 840,000
Direct labor	168,000
Manufacturing overhead	672,000
Total product cost	$1,680,000

2. Product Cost per Unit = Total Product Cost/Number of Units

Product Cost per Unit = $1,680,000/2,750 units = $610.91

Solution

Product Cost in Total and Per Unit

Grin Company manufactures digital cameras. In January, Grin produced 2,750 cameras with the following costs:

Direct materials	$840,000
Direct labor	168,000
Manufacturing overhead	672,000

There were no beginning or ending inventories of WIP.

Required:

1. What was the total product cost in January?

$ 1680000

2. What was the product cost per unit in January? **Round your answer to the nearest cent.**

$ 610.91 per unit

21. EX.02.52.ALGO (Algorithmic)

Prime Cost and Conversion Cost

Grin Company manufactures digital cameras. In January, Grin produced 3,000 cameras with the following costs:

Direct materials	$1,000,000
Direct labor	136,000
Manufacturing overhead	416,000

There were no beginning or ending inventories of WIP.

Required:

If required, round your answers to the nearest cent.

1. What was the total prime cost in January?

$ ✓

1136000

2. What was the prime cost per unit in January?

$ ✓

378.67

per unit

3. What was the total conversion cost in January?

$ ✓

552000

4. What was the conversion cost per unit in January?

$ ✓

184.00

per unit

Feedback

Check My Work

1. Prime costs consist of direct materials and direct labor.

2. Divide total prime costs by the number of units produced.

3. Conversion cost is the sum of direct labor and manufacturing overhead.

4. Divide total conversion costs by the number of units produced.

Post-Submission

1.

Direct materials	$1,000,000
Direct labor	136,000
Total prime cost	$1,136,000

2. Prime Cost per Unit = Total Prime Cost/Number of Units

Prime Cost per Unit = $1,136,000/3,000 units = $378.67

3.

Direct materials	$136,000
Manufacturing overhead	416,000
Total conversion cost	$552,000

4. Conversion Cost per Unit = Total Conversion Cost/Number of Units

Conversion Cost per Unit = $552,000/3,000 units = $184.00

Solution

Prime Cost and Conversion Cost

Grin Company manufactures digital cameras. In January, Grin produced 3,000 cameras with the following costs:

Direct materials	$1,000,000
Direct labor	136,000
Manufacturing overhead	416,000

There were no beginning or ending inventories of WIP.

Required:

If required, round your answers to the nearest cent.

1. What was the total prime cost in January?

$ 1136000

2. What was the prime cost per unit in January?

$ 378.67 per unit

3. What was the total conversion cost in January?

$ 552000

4. What was the conversion cost per unit in January?

$ 184.00 per unit

22. EX.02.55.ALGO (Algorithmic)

Direct Materials Used, Cost of Goods Manufactured

In September, Lauren Ashley Company purchased materials costing $180,000 and incurred direct labor cost of $110,000. Overhead totaled $345,000 for the month. Information on inventories was as follows:

	September 1	**September 30**
Materials	$100,000	$120,000
Work in process	$80,000	$70,000
Finished goods	$80,000	$70,000

1. What was the cost of direct materials used in September?

$ ✓

160,000

2. What was the total manufacturing cost in September?

$ ✓

615,000

3. What was the cost of goods manufactured for September?

$ ✓

625,000

4. Assume that Lauren Ashley Company's monthly incurred direct labor cost increased by 25% and total overhead costs decreased by 20%. Using Excel (or some other spreadsheet software tool), calculate Lauren Ashley's new cost of goods manufactured that results from the changes in direct labor cost and overhead costs.

$ ✓

583,500

Even for a relatively simple exercise, this requirement illustrates the time and effort savings of utilizing technology in setting up and solving formulas as typically **required** ✓ in management accounting data analytic settings.

Feedback

Check My Work

1. Direct materials used = Beginning materials + Purchases - Ending materials.

2. Total manufacturing cost = Direct materials used + Direct labor + Overhead.

3. The cost of goods manufactured = Beginning WIP + Total manufacturing cost - Ending WIP.

Post-Submission

1.

Materials inventory, September 1	$100,000
Materials purchases in September	180,000
Materials inventory, September 30	(120,000)
Direct materials used in September	$160,000

2.

Direct materials	$160,000
Direct labor	110,000
Manufacturing overhead	345,000
Total manufacturing cost	$615,000

3.

Total manufacturing cost	$615,000
Add: Work in process, September 1	80,000
Less: Work in process, September 30	(70,000)
Cost of goods manufactured	$625,000

4. Students are instructed to use a spreadsheet software tool, such as Excel, to help them set up and solve the exercise’s requirements. For requirement 4, the spreadsheet needs to be adjusted to accommodate the two cost changes. Most students should be able to simply type in the new increased total direct labor cost and new decreased total overhead costs and easily obtain the resulting new cost of goods manufactured. However, requirement 4's sensitivity analysis requires more time and effort for students who instead elect to write out and solve these requirements by hand. Therefore, even for a relatively simple exercise, this requirement illustrates the time and effort savings of utilizing technology in setting up and solving formulas as typically required in management accounting data analytic settings.

As shown in detail below, total direct labor increases by 25% from $110,000 to $137,500, and total overhead decreases by 20% from $345,000 to $276,000. As a result, the new cost of goods manufactured decreases by $41,500 from a total of $625,000 to a total of $583,500.

Materials inventory, September 1	$100,000
Materials purchases in September	180,000
Materials inventory, September 30	(120,000)
Direct materials used in September	$160,000

Direct materials	$160,000
Direct labor	137,500*
Manufacturing overhead	276,000**
Total manufacturing cost	$573,500

Total manufacturing cost	$573,500
Add: Work in process, September 1	80,000
Less: Work in process, September 30	(70,000)
Cost of goods manufactured	$583,500

* $110,000 x 1.25

** $345,000 x 0.8

Solution

Direct Materials Used, Cost of Goods Manufactured

In September, Lauren Ashley Company purchased materials costing $180,000 and incurred direct labor cost of $110,000. Overhead totaled $345,000 for the month. Information on inventories was as follows:

	September 1	**September 30**
Materials	$100,000	$120,000
Work in process	$80,000	$70,000
Finished goods	$80,000	$70,000

1. What was the cost of direct materials used in September?

$ 160,000

2. What was the total manufacturing cost in September?

$ 615,000

3. What was the cost of goods manufactured for September?

$ 625,000

4. Assume that Lauren Ashley Company's monthly incurred direct labor cost increased by 25% and total overhead costs decreased by 20%. Using Excel (or some other spreadsheet software tool), calculate Lauren Ashley's new cost of goods manufactured that results from the changes in direct labor cost and overhead costs.

$ 583,500

Even for a relatively simple exercise, this requirement illustrates the time and effort savings of utilizing technology in setting up and solving formulas as typically required in management accounting data analytic settings.

23. EX.02.57.ALGO (Algorithmic)

Cost of Goods Sold, Sales Revenue, Income Statement

Jasper Company provided the following information for last year:

Sales in units	280,000
Selling price	$12
Direct materials	200,000
Direct labor	545,000
Manufacturing overhead	110,000
Selling expense	437,000
Administrative expense	854,000

Last year, beginning and ending inventories of work in process and finished goods equaled zero.

Required:

Calculate the cost of goods sold for last year.

$ ✓

855,000

Feedback

Check My Work

Add direct materials, direct labor, and manufacturing overhead for total cost of goods sold. There were no beginning or ending inventory balances.

Post-Submission

Direct materials	$200,000
Direct labor	545,000
Manufacturing overhead	110,000
Cost of goods sold	$855,000

Note: Because there were no beginning or ending work-in-process or finished goods inventories, there is no difference here between cost of goods sold and cost of goods manufactured, so no interim calculations for them are necessary.

Cost of Goods Sold, Sales Revenue, Income Statement

Jasper Company provided the following information for last year:

Sales in units	280,000
Selling price	$12
Direct materials	200,000
Direct labor	545,000
Manufacturing overhead	110,000
Selling expense	437,000
Administrative expense	854,000

Last year, beginning and ending inventories of work in process and finished goods equaled zero.

Required:

Calculate the cost of goods sold for last year.

$ 855,000